

Making HTTP header injection critical via response queue poisoning

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Content

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[image: image]

HTTP header injection is often under-estimated and misclassified as a moderate severity flaw equivalent to [XSS](#) or worse, Open Redirection. In this post, I'll share a simple technique I used to take a header injection vulnerability, make it critical, and earn a \$12,500 bounty.

This technique applies to both request header injection on front-end servers, and response header injection on back-end servers.

Background

This all started when out of the blue, a stranger emailed me a path-based request header injection and asked if I had any ideas for exploitation. The vulnerability was on a major, high-traffic site serving critical functionality that we'll refer to as 'redacted.net':

```
`GET /%20HTTP/1.1%0d%0aHost:%20redacted.net%0d%0a%0d%0a HTTP/1.1 Host: redacted.net
HTTP/1.1 200 OK
```

```
GET /%20HTTP/1.1%0d%0anothost:%20redacted.net%0d%0a%0d%0a HTTP/1.1 Host: redacted.net
HTTP/1.1 400 Bad Request`
```

I don't typically engage with emails like this, as usually the reporter has got stuck because it's genuinely unexploitable and I don't have any tricks up my sleeve to help. However, I'd [long suspected](#) that it might be possible to upgrade header injection vulnerabilities into request smuggling. Also, the target website was under a bug bounty program which is known for competitive bounty payouts, and the reporter - [xorb](#) - agreed to a 50/50 bounty split if I could help.

Upgrading header injection into HTTP request smuggling

The concept is simple - you can convert a request header injection into a more serious HTTP desync with a few easy steps.

First, identify where your injection is occurring and add anything necessary to cleanly exit the context:

```
`GET /%20HTTP/1.1%0d%0a%0d%0a HTTP/1.1
HTTP/1.1 400 Bad Request Connection: close`
```

Then inject essential headers to ensure the back-end keeps the connection open after responding to the initial request:

```
`GET /%20HTTP/1.1%0d%0aHost:%20redacted.net%0d%0aConnection:%20keep-alive%0d%0a%0d%0a
HTTP/1.1
HTTP/1.1 200 OK Connection: keep-alive`
```

At this point we can specify a second request fully under our control, so we're set up for a classic request smuggling attack. The only significant difference is that we'll need to account for the server appending additional headers/body after our injection. Here's two of the many options for cross-user exploitation.

Specifying a malicious prefix to poison either the next user's request, or a web cache:

```
GET /%20HTTP/1.1%0d%0aHost:%20redacted.net%0d%0aConnection:%20keep-
alive%0d%0a%0d%0aGET%20/redirect%20HTTP/1.1%0d%0aHost:%20astify.com%0d%0a%0d%0aContent-
Length:%2050%0d%0a%0d%0a HTTP/1.1
```

Or crafting our prefix to combine with the trailing junk and create a complete second request in order to trigger response queue poisoning.

```
GET /%20HTTP/1.1%0d%0aHost:%20redacted.net%0d%0aConnection:%20keep-  
alive%0d%0a%0d%0aGET%20/%20HTTP/1.1%0d%0aFoo:%20bar HTTP/1.1
```

I went for the latter option, which successfully lead to me intermittently receiving responses intended for other authenticated users. I have a beautiful screenshot showing this, but sadly I was unable to get permission to name the target.

This was sufficient to prove critical impact to the target, who patched it in under 24 hours and awarded a \$12,500 bounty.

If you run into issues applying this technique for yourself, these two closely related posts may be useful:

- [Response queue poisoning in Jira](#)
- [HTTP request smuggling using CRLF injection](#)

Response header injection and the stacked-response problem

As we've seen, upgrading request header injection into a desync is pretty easy. Sometimes, upgrading response header injection is similarly straightforward. However, other times it mysteriously fails. I recently discovered a defence mechanism which I believe explains this, and hints at a possible solution.

When web browsers read in a response, if they encounter more data than the server promised in the Content-Length header, they truncate the response and close the connection. I dubbed this the [stacked-response problem](#), and found it made exploiting Client-Side Desync vulnerabilities tougher but not impossible.

I now suspect some major front-end servers have a similar mechanism, which has two security implications:

- Regular desync attacks are unaffected, but response-queue poisoning is mitigated
- It's difficult to convert response header injection into a HTTP desync

If your attempts at causing a desync via response header injection fail, you may have encountered this mechanism. To bypass it, you need to delay the injected response so that the front-end's over-read doesn't see it.

One possible approach for this is to inject a large number of newlines, which are typically consumed by servers without triggering request/response processing.

Ultimately, this aspect needs further research. If you encounter this challenge on a bug bounty program and get stuck, I'd be happy to see if I can help. I should also mention if the website you've found header injection on doesn't have a front-end, these techniques won't work as-is but you may still be able to achieve a client-side desync.

Final notes

I suspect these techniques used to be known but got forgotten alongside [HTTP Request Smuggling](#), which explains why some people refer to response header injection as 'response splitting' even though they never actually split the response. For a deeper exploration of the phenomenon of forgotten security knowledge, check out [Hunting Evasive Vulnerabilities](#).

I hope these techniques are useful for you, we'd [love to hear](#) if you find success with them.

[Request Smuggling](#)

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